Infinite Series

If (a_n) is an infinite sequence of numbers, then the formal expression $a_1+a_2+\cdots$ is called an ***infinite series*** and is denoted as $\sum_{k=1}^{\infty} a_k$.

In the beginning of the section we have seen the infinite series

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots$$

and intuitively felt that "*its sum*" is 1 (in the cake problem). Actually in the cake problem, the stage by stage availability of the cake in the plate B is given in the following sequence.

$$\frac{1}{2},\ \frac{1}{2}+\frac{1}{4},\ \frac{1}{2}+\frac{1}{4}+\frac{1}{8},\ \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}\ \cdots$$

The nth term of this sequence is

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots+\frac{1}{2^n}$$

which is equal to $\frac{2^n-1}{2^n}$. If s_n denote this sum, then $\lim_{n\to\infty} s_n = 1$. This is one of the reasons for us to feel that the sum

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots \qquad \text{is 1.}$$

Motivated by this we may define the sum of infinitely many numbers. Let (a_n) be a sequence of real numbers and let $s_n = a_1 + a_2 + a_3 + \cdots + a_n$. If the sequence (s_n) converges to a limit s, then it is meaningful to say that the sequence (a_n) is "*summable*" and the sum is s. In this case, we write

$$a_1 + a_2 + a_3 + \cdots = s$$

It is customary to say that the series converges to s.

Definition 5.6

Let $\sum_{n=1}^{\infty} a_n$ be a series of real numbers and let

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n, n \in \mathbb{N}$$

The sequence (s_n) is called the ***partial sum sequence*** of $\sum_{n=1}^{\infty} a_n$. If (s_n) converges and if $\lim_{n\to\infty} s_n = s$, then the series is said to be a ***convergent series*** and s is called the ***sum*** of the series.

We write $\sum_{n=1}^{\infty} a_n = s$. Let us see some examples. The series $\sum_{n=1}^{\infty} (-1)^{n+1}$ does not converge because the partial sum sequence $1, 0, 1, 0, 1, 0, \ldots$ does not converge.



We cannot apply algebraic rules meant for finite series to an infinite series blindly. Consider $\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \cdots$ If $S = 1 - 1 + 1 - 1 + \cdots$ then one may argue that $S = 0$ or 1 or $\frac{1}{2}$ according to $S = (1 - 1) + (1 - 1) + \cdots$, $S = 1 + (-1 + 1) + (-1 + 1) + \cdots$ or $1 - S = 1 - (+1 - 1 + 1 - 1 + \cdots = S)$ respectively.